

CH3.1 多項式函數

1. -1 2. -107 3. (1)(2, $-5, 6, -7$) (2) -7.06

4. $(-5, -16)$ 5. $-4x + 5$ 6. -24 7. (C)(D)

8. $(\frac{1}{2}, 2)$ 9. -16 10. $k < \frac{1}{2}$

解析

1. $(x^3 + ax^2 + 3x - 1)(x^3 - 4x^2 + 3x + 5)$ 乘開後

x^4 項為 $(x^3)(3x) + (ax^2)(-4x^2) + (3x)(x^3)$

$$= (3 - 4a + 3)x^4$$

$$\Rightarrow 6 - 4a = 10 \Rightarrow a = \underline{-1}$$

2. $f(x) = x^5 - 9x^4 + 13x^3 + 32x + 12$

求 $f(7) \Rightarrow$ 可視為 $f(x)$ 除以 $x - 7$ 的餘式

故由綜合除法

$$\begin{array}{r|rrrrrr} 1 & -9 & +13 & +0 & +32 & +12 \\ & +7 & -14 & -7 & -49 & -119 \\ \hline 1 & -2 & -1 & -7 & -17 & -107 \end{array} \Bigg| 7$$

$$\therefore f(7) = -107$$

$$\begin{array}{r|rrrr} 54 & -99 & +66 & -20 \\ & +18 & -27 & +13 \\ \hline 54 & -81 & +39 & \\ \hline 18 & -27 & +13 & \\ & +6 & -7 & \\ \hline 18 & -21 & +6 & \\ \hline 6 & -7 & & \\ & +2 & & \\ \hline 6 & -5 & & \\ \hline a=2 \end{array} \Bigg| \frac{1}{3}$$

3. (1) 計算如右

$$\begin{aligned}\therefore f(x) &= 2(3x-1)^3 - \\ &\quad 5(3x-1)^2 + \\ &\quad 6(3x-1) - 7\end{aligned}$$

故序組 (a, b, c, d)

$$= (2, -5, 6, -7)$$

$$(2) f(0.33) = 2(0.99-1)^3 - 5(0.99-1)^2 + 6(0.99-1) - 7 = -0.000002 - 0.0005 - 0.06 - 7 \approx -7.06$$

$$4. \text{ 設 } f(x) = (x^3 + 8)Q(x) + 3x^2 - 2x - 1$$

$$\begin{aligned}\Rightarrow xf(x) &= x(x^3 + 8)Q(x) + 3x^3 - 2x^2 - x \\ &= x(x+2)(x^2 - 2x + 4)Q(x) + (x^2 - 2x + 4)(3x + 4) - 5x \\ &\quad - 16 = (x^2 - 2x + 4)[x(x+2)Q(x) + 3x + 4] - 5x - 16\end{aligned}$$

$\therefore$ 餘式為 $-5x - 16$

即數對 $(a, b) = (-5, -16)$

$$5. \text{ 設 } f(x) \text{ 除以 } x^2 + x - 2 \text{ 的餘式為 } ax + b$$

$$\text{則 } f(x) = x^{10} + 2x^9 - 5x + 3 = (x^2 + x - 2)Q(x) + ax + b = (x+2)(x-1)Q(x) + ax + b$$

$$\begin{cases} f(-2) = 1024 - 1024 + 10 + 3 = -2a + b \\ f(1) = 1 + 2 - 5 + 3 = a + b \end{cases}$$

$$\Rightarrow \begin{cases} -2a + b = 13 \\ a + b = 1 \end{cases} \Rightarrow \begin{cases} a = -4 \\ b = 5 \end{cases}, \text{ 故餘式為 } \underline{-4x + 5}$$

$$6. \text{ 令 } f(x) = a(x-1)(x-2)(x-3),$$

$$\text{且 } f(4) = a(4-1)(4-2)(4-3) = 6 \Rightarrow a = 1$$

$\therefore f(x) = (x-1)(x-2)(x-3)$ 除以 $x+1$ 之餘式為

$$f(-1) = (-1-1)(-1-2)(-1-3) = \underline{-24}$$

7. (A) $\times$: 開口向下, $a < 0$

(B) $\times$: 過(0, c)作切線, 右下降, 斜率 $< 0, b < 0$

(C) $\circ$: 點 (0, c) 在 x 軸上方, $c > 0$

(D) $\because$ 與 x 軸有二交點, $D > 0$

(E) X : 對稱軸為 $x = -1 \Rightarrow f(-3) = f(1) < 0, 9a - 3b + c < 0$ 故選(C)(D)

$$8. y = (x + 2)^2 + 12 = ax^2 + 4ax + 4a + 12$$

$$\Rightarrow \begin{cases} 4a = b \dots\dots\dots ① \\ 4a + 12 = \frac{7}{a} \dots\dots ② \end{cases}$$

$$\text{由②得 } 4a^2 + 12a - 7 = 0 \Rightarrow a = \frac{1}{2} \text{ 或 } -\frac{7}{2}$$

因開口向上, 所以取 $a = \frac{1}{2}$ 代入①得 $b = 2$

$$\text{故數對 } (a, b) = \left(\frac{1}{2}, 2\right)$$

$$9. \because f(x) \text{ 滿足 } f(2 - t) = f(-4 + t)$$

$$\therefore f(2) = f(-4) \text{ , 且 } f(3) = f(-5)$$

$$\Rightarrow f(-5) - f(-4) = f(3) - f(2) = -16$$

$$10. 2x^2 + x - k + 1 > 3x \text{ 恆成立}$$

$$\Rightarrow 2x^2 - 2x + (1 - k) > 0 \text{ 恆成立}$$

$$\text{判別式} = 4 - 8(1 - k) < 0 \Rightarrow k < \frac{1}{2}$$